

Performance Characteristics (continued)

Trigger type selections

	EDUX1052A/EDUX1052G	DSOX1202A/DSOX1202G DSOX1204A/DSOX1204G
Edge	Trigger on a rising, falling, alternating or either edge of any source	
Pattern/state	Not available	Trigger when a specified pattern/state on any combination inputs is entered ⁹
Pulse width	Trigger on a pulse of a selected channel with a time duration that is 'less than a value,' 'greater than a value' or 'inside a time range' Range minimum: 10 ns, 10 s max	
Setup and hold	Not available	Trigger and clock/data setup and/or hold time violation. Setup time can be set from –7 ns to 10 s. Hold time can be set from 0 s to 10 ns
Rise/fall time	Not available	Trigger on rise-time or fall-time edge-speed violations (< or >) based on a user-selectable threshold and time setting range between 5 ns and 10 s
Video	Trigger on all lines or individual lines; odd/even or all fields from the composite video; or broadcast standards (NTSC, PAL, SECAM, and PAM-M)	
I ² C	Trigger at a start/stop condition or user-defined frame with address and/or data values. Also, trigger on missing acknowledge, restart, EEPROM read and 10-bit write	
RS-232/422/485/UART	Trigger on Rx or Tx start bit, stop bit, data content or parity error	
SPI	Not available	Trigger on SPI (Serial Peripheral Interface) data pattern during a specific framing period. Supports positive and negative chip select framing as well as clock idle framing. Supports MOSI or MISO (4-channel models) data as half duplex data
CAN	Not available	Trigger on CAN (controller area network) version 2.0A and 2.0B signals. Trigger on the start of frame (SOF) bit, remote transfer request frame ID (RTR), data frame ID (~RTR), remote or data frame ID, data frame ID + data, error frame, all errors, acknowledge error, or overload frame.
LIN	Not available	Trigger on LIN (Local Interconnect Network) sync break, frame ID, frame ID + data, parity error, or checksum error

9. The pattern must have stabilized for a minimum of 5 ns to qualify as a valid trigger condition.